

# Ruzicka similarity-based brain EEG clustering for improved intelligent epilepsy diagnosis

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## ARTICLE INFO

### Keywords:

Feature selection  
Clustering  
Particle swarm optimization (PSO)  
Ruzicka clustering  
EEG Signals  
Machine Learning Classifiers

## ABSTRACT

This paper aims to introduce a novel clustering method for electroencephalogram (EEG) based on Ruzicka mathematical similarity and incorporates Particle Swarm Optimization (PSO) to enhance feature selection. Medical datasets often contain both convergent and divergent features, making feature selection a crucial step for accurate disease diagnosis and public health applications. The proposed Ruzicka-based clustering method groups EEG records into non-overlapping subgroups according to a defined similarity metric. Cluster centers are determined using a polynomial-based calculation, after which EEG records are assigned to clusters based on the Ruzicka similarity measure. After clustering the EEG records into highly coherent groups, PSO algorithm is employed to identify the most effective subset of features. This process enhances classification accuracy and contributes to more reliable diagnostic outcomes by combining clustering with feature selection. The selected features are then evaluated using multiple classifiers, including Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), K-Nearest Neighbors (KNN), and Naive Bayes (NB). Accuracy, recall, f1-score and precision measures are conducted to evaluate the model's performance. Experimental validation is carried out on the Bonn University EEG dataset. With both RF and NB classifiers, the proposed model has achieved up to 100% accuracy compared to other models. The proposed method can be implemented in medical organizations as a decision-support system to assist healthcare professionals in analyzing EEG patterns. Its integration can enhance the accuracy and efficiency of disease diagnosis, leading to improved patient care.

## 1. Introduction

Approximately fifty million people worldwide have suffered from epilepsy in various age groups [1]. Traditionally, doctors who specialize in brain tissue diseases rely on visual examination methods. Technically, this approach is prone to human error, costly, and difficult [2]. EEG is used to detect and monitor epilepsy seizures. A brain activity monitor records electrical activity in the brain cells [3]. EEG is a relatively inexpensive, non-invasive diagnostic tool and is used to detect diseases associated with brain cell activity such as issues with sleep, epileptic seizure, and Alzheimer's [4]. However, EEG signal contains a lot of incomprehensible information, making it inappropriate for precise medical detection [5]. Furthermore, complexity and noise are major limitations that affect accurate classification of EEG [6,7].

With the aim of select important features, feature selection or

optimization algorithms are applied. Generally, efficient classification considers only effective features. It mainly supports correct decision-making. An analytical and valuable approach is aimed to group data that are not associated with labels into groups by clustering. It obtains important information from the dataset [8]. As mentioned in [9], data in the same cluster are distinguished by highest similarity and greatest variation among different clusters. Additionally, it has enhanced the PSO algorithm [10].

Machine learning (ML) techniques are used for analyzing health-related data. Neurologists using machine-learning applications have improved diagnostic and treatment capabilities [11]. PSO is a common algorithm that can handle variety of optimization issues, slow convergence and local optimal trapping are two primary problems that PSO confronts [12]. Consequently, performance is impaired [13]. This optimizer frequently needs more feature selection, further emphasizing that

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<https://doi.org/10.1016/j.cmpbup.2025.100229>

it takes more time to carry out. Then, execution of optimizer has increased the complication. Genetic algorithm (GA), Ant Colony Optimization (ACO) and PSO algorithm have been derive behavior from biological activity. Then, these algorithms adopted in a various research domains, such as energy system, robotics, engineering for aircraft and artificial intelligence [14]. This study seeks to demonstrate how to improve EEG signal clustering using Ruzicka similarity to support PSO optimization algorithm select essential features for epileptic seizure identification.

Fig. 1 illustrates the methodology of EEG signal clustering for early epileptic-seizure detection. The discussion starts by outlining the EEG methodology and recordings, followed by preprocessing steps designed to remove noise and retain useful information. Then goes on the process of feature selection focusing on removing (discarding) irrelevant features. This section ends with the use of ML classifiers for classification.

### 1.1. Motivation and problem statement

Besides their real potential for the detection of neural disorders related to brain cell activity [15], EEG data analysis has attracted considerable attention in recent research. However, EEG datasets are bathing in high dimensionality and impose significant hindrance on the use of traditional ML-based algorithms [16,17]. Some neurologists recently resorted to a subjective visual analysis of EEG signals. While efficient, this is a time-consuming and labor-intensive task. Besides, patients who are oftentimes in rural areas have difficulty gaining timely access to healthcare check-ups. Such problems therefore accentuate the need to improve the accuracy of ML-based classification techniques that might eventually work toward improving the efficacies of diagnostics and their availability.

### 1.2. Main contributions

A new clustering scheme has been developed and implemented based on Ruzicka similarity. The method is intended for preprocessing an EEG analysis to better the efficiency of classification performance. The clustering methods carry out the task of identifying features that show better internal concinnity all across the dataset to improve the efficiency of the feature selection process. Effectively creating a harmonizing environment for feature selection approaches and optimization methods like PSO, this technique has produced very good results for epileptic seizure detection [18]. Also, with the proposed Ruzicka-PSO clustering

method, the probability of stagnation will be decreased, thus improving the exploratory process.

### 1.3. Evaluation strategy and setup

The dataset of Bonn university is used for verifying the model proposed. There are five different folders that contain a total of 100 records, each with 4097 attributes. The EEG signals that are used in the study last for 23.6 s. The dataset is divided into two clusters, such that the weights of all records are calculated in a polynomial equation and ranked accordingly. Cluster centers are determined based on the number of vectors, clusters, and calculated displacements. Ruzicka similarity for each EEG signal is computed relative to these cluster centers. Each cluster represents a subset of the dataset from which PSO algorithm is used to select significant features. These features are subsequently input into the classifier.

### 1.4. Paper organization

The rest of this paper is arranged into sections. Section 2 illustrates related work. Section 3 depicts Ruzicka similarity mathematically. Section 4 illustrates the proposed system. Section 5 shows the computational and comparative results of the suggested method with other clustering methods. Lastly, conclusion of the overall paper and future work are summarized in Section 6.

## 2. Related work

The analysis of EEG signals for identifying and classifying epileptic seizures represents an important research domain. In this section, we summarize existing works that propose different models to improve the classification of EEG signals.

Alalayah et al. [19]. proposed a model for epileptic seizure for EEG detection. Discrete Wavelet Transform (DWT) was employed to extract relevant frequency and time information from EEG signals, establishing its suitability for EEG analysis. t-distributed Stochastic Neighbor Embedding (t-SNE) and Principal Component Analysis (PCA) have been implemented for decreasing data dimensionality, while data points are grouped using K-means clustering for improved classification. Classification algorithms, including DT, RF, KNN, Multilayer Perceptron (MLP) and Extreme Gradient Boosting (XGBoost) are evaluated using accuracy, precision, recall, and F1-score. Dataset sourced from the University of

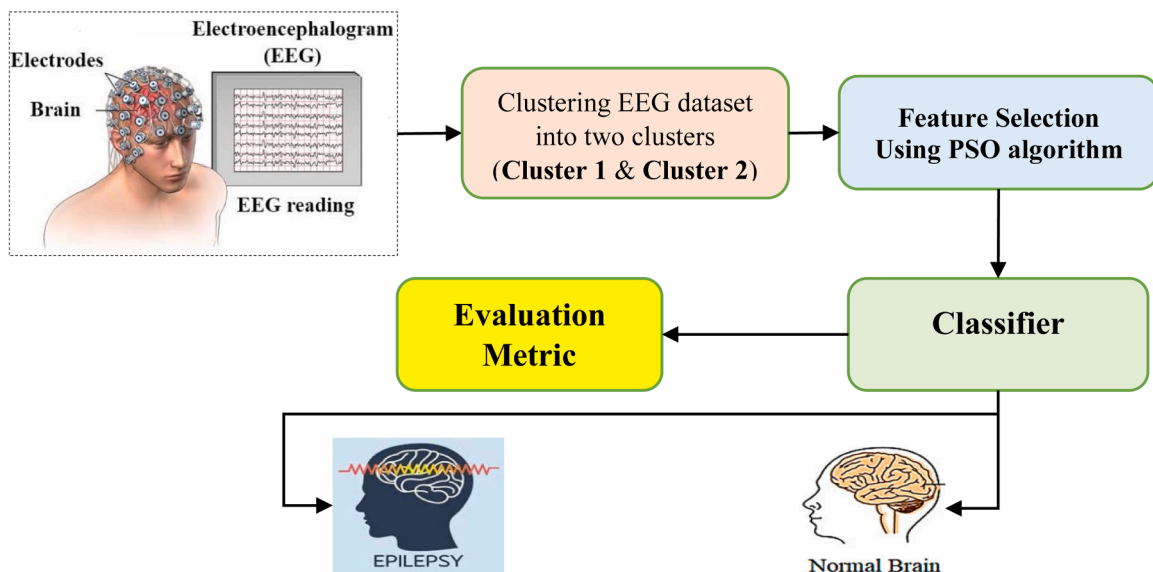


Fig. 1. Methodology of EEG Clustering for Early Detection of Epileptic Seizures.

Cape Town (UCT), divided into 80 % training and 20 % testing. The accuracy performance metrics include 97.96 % (DWT + PCA + RF) and 98.98 % (K-means + PCA + MLP). Furthermore, the study does not consider processing time and computation complexity, which limits practical applicability.

Al-Shammary et al. [20], proposed a technique of clustering EEG data into clusters with the use of Minkowski distance such that their novel scheme was free from the stagnation problems found in conventional classification approaches. This had been regarded as a crucial area since optimization algorithm problems such as local stalling and inconsistent results were often misclassified when dealing with high-dimensional EEG datasets. Important features were selected using PSO, while classification models include RF, KNN, SVM, DT, and NB. The experimental baseline shows that statements in the Bonn University dataset achieved 100 percent accuracy (in the case of binary classes) when validated through RF and NB classifiers. But when feature dimensions led to high increases in KNN performance decreased and there was a dichotomy in performance as classes grew.

Prabhakar and Won [21] highlighted the need to automate seizure diagnosis by combining clustering techniques with ML classifiers. Their study employed both learning-based techniques (fuzzy c-means, k-means) and bio-inspired clustering (cuckoo search and Dragonfly clusters) to improve feature selection on the Bonn University dataset. The evaluation revealed varying performances across classifiers, with SVM paired with cuckoo search achieving the highest accuracy of 99.48 %. However, the use of Dragonfly clusters with KNN yielded minimal accuracy (75.5 %). Notably, their analysis lacked critical considerations of time efficiency, which limit its practical application.

Li et al. [22], have focused on clustering unlabeled EEG signals and measuring similarity between data points. The proposed model has presented EEG clustering method. ShVEEGc is applied to improve the shapely value of collaborative game theory. It has measured correlations between data points of EEG using improved cosine similarity. ShVEEGc has avoided the computation center of cluster as in traditional methods like K-means clustering. Technically, it relies on internal relationship between EEG signals to construct clusters stably. Kappa (K), Rand Index (RI), F1-score, and Normalized Mutual Information (NMI) metrics are utilized for assessing ShVEEGc performance. Moreover, several metrics are applied to measure the similarity of EEG, including Category Difference, Discrimination Ability (DA), Category Difference (CD), and Composite indicator (CI). Eleven dataset are conducted for evaluating the proposed model, bci\_I\_a, bci\_I\_b, bciII\_a and I\_b, bciIII\_V\_s1, bciIII\_V\_s2, bciIII\_V\_s3, bci\_2a\_s1, bci\_2b\_s2, bci\_2a\_s2, bci\_2a\_s3, bci\_2b\_s1 and bci\_2b\_s3 are used. ShVEEGc is proven for outperformance and effectiveness compared to other methods. The computational complexity of ShVEEGc is increased with high-dimensional data, making it not practical for large EEG dataset.

Du et al. [23], addressed challenges in unsupervised clustering and supervised classification for epileptic EEG analysis. Their model combined noise reduction techniques, such as Complete Ensemble Empirical Mode Decomposition (CEEMDAN) and Continuous Wavelet Transform (CWT), with advanced clustering using the Sparrow Search Algorithm

(SSA) to optimize DBSCAN. While their model achieved impressive results on clustering metrics such as Silhouette Coefficient (SC) and Calinski-Harabasz Score (CH), it failed to address computational time and complexity issues, limiting its clinical applicability.

Previous studies generally show major limitations in terms of expensive computation cost, not being able to discern optimum cluster centers, and lower accuracy in handling more complex datasets comprised of highly similar EEG records. The uniqueness of the present work is using the Ruzicka similarity measure for clustering high similarity EEG signals and combining with feature selection method (PSO). This integration reduces dimensionality while maintaining both accuracy and efficiency, thereby addressing the shortcomings of existing approaches and offering a more adaptable solution for epilepsy diagnosis. Table 1 summarizes the previous studies, including their datasets, main findings, and drawbacks.

### 2.1. Ruzicka similarity measure

It is a measure employed to compute the similarity between two vectors, often in the context of feature selection, clustering, or classification tasks. Ruzicka similarity aims to distribute data between clusters depends on connection and irrelevance among them. It is a similarity between two vectors  $\mathbf{X} = [x_1, x_2, \dots, x_n]$  and  $\mathbf{Y} = [y_1, y_2, \dots, y_n]$  can be defined by Eq.1:

$$\text{Ruzicka Similarity} = \frac{\sum_{i=1}^n \min \{x_i, y_i\}}{\sum_{i=1}^n \max \{x_i, y_i\}} \quad (1)$$

Where  $n$  is the number of features, and both  $\mathbf{X}$  and  $\mathbf{Y}$  represent two vectors that are the same size and belong to the real numbers group.

For non-negative feature vectors, this measure is mathematically equivalent to the continuous Jaccard index, which generalizes the well-known Jaccard coefficient from binary sets to real-valued data. In both cases, similarity is quantified as the overlap between two vectors relative to their total magnitude, producing values bounded between 0 and 1. This property makes the metric highly interpretable, as values approaching 1 indicate a strong similarity while values close to 0 indicate little or no similarity. The reason for using Ruzicka instead of using other methods like Cosine and Euclidean distance for similarity measurement is because it is much favorable towards EEG features. For example, while the Euclidean distance is very sensitive to absolute scale and shines variance in the amplitude, the Ruzicka similarity supports invariance under positive rescaling, thereby not changing the similarity value if both vectors are multiplied by the same factor. While cosine similarity measures the angular alignment, Ruzicka captures the fraction of magnitudes common to each in a manner that more accurately reflects a structural overlap in EEG descriptors. By these features, Ruzicka makes the clustering of EEG signals much more appropriate, as records will cluster EEG signals according to their strongest shared features before the feature selection stage.

**Table 1**  
Summary of Previous Studies.

| Ref.                    | Proposed Technique                                      | Dataset                          | Findings                                                                 | Drawbacks                                                        |
|-------------------------|---------------------------------------------------------|----------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------|
| Alayah et al. [19]      | Model for epileptic seizure detection using EEG signals | UCT(training 80 %, testing 20 %) | Accuracy: (DWT+PCA+RF) 97.96 %, (K-means + PCA + MLP) 98.98 %            | Did not account for time processing or computational complexity. |
| Al-Shammary et al. [20] | Clustering strategy with Minkowski distance             | University of Bonn dataset       | Accuracy: (RF+NB) 100 %                                                  | Negative impact of increasing                                    |
| Prabhakar and Won [21]  | Combining clustering techniques with ML classifiers     | University of Bonn dataset       | Accuracy: (SVM+cuckoo search) 99.48 %, (Dragonfly clusters + KNN) 75.5 % | Lacked consideration of time efficiency                          |
| Li et al. [22]          | ShVEEGC Clustering with Cosine similarity               | Eleven EEG datasets              | Provides high recall and F1-Score                                        | High computational complexity limits scalability.                |
| Du et al. [23]          | Clustering and classification for EEG analysis.         | University of Bonn (Germany)     | Achieved high SC and CH scores                                           | Failed to address computational time and complexity.             |

### 3. Proposed methodology

This work proposes an efficient clustering method based on the Ruzicka similarity metric to support the PSO in identifying the most relevant features from EEG. The Ruzicka similarity measures the overlap between two feature vectors, helping to distinguish between seizure EEG and healthy EEG patterns. In this work, the number of clusters is intentionally fixed at two, and this is adopted as a simplifying assumption for the proposed framework. The formation of two clusters provides a unified framework for developing the sub-datasets in terms of feature selection and allows PSO to search in regions of greater similarity. This allows for more robust selection of relevant features and improved classification accuracy. Although it was thought to have partitioned into more than two clusters, such division will ultimately lead to the

fragmentation of the data from the intended harmony and will diminish the effect of feature selection. Hence, the two-cluster scheme is the best balance of simplicity, interpretability, and robustness.

The technique emphasizes the identification of similar EEG records for creating clusters in the high-dimensional datasets. This clustering method contributes to enhanced exploration and decreases the risk of getting stagnation in PSO. A polynomial-based method is employed to find the centers of clusters after which the dataset is separated into two clusters, as illustrated in Fig. 2. However, conventional techniques used in determining the centers of clusters are generally unpredictable and dependent on heuristics [24]. Conversely, the polynomial method assigns deterministic weights to each vector by putting together feature values and positional indices, sorts in ascending order, and incorporates the selected most representative vectors as cluster centers. This gives the

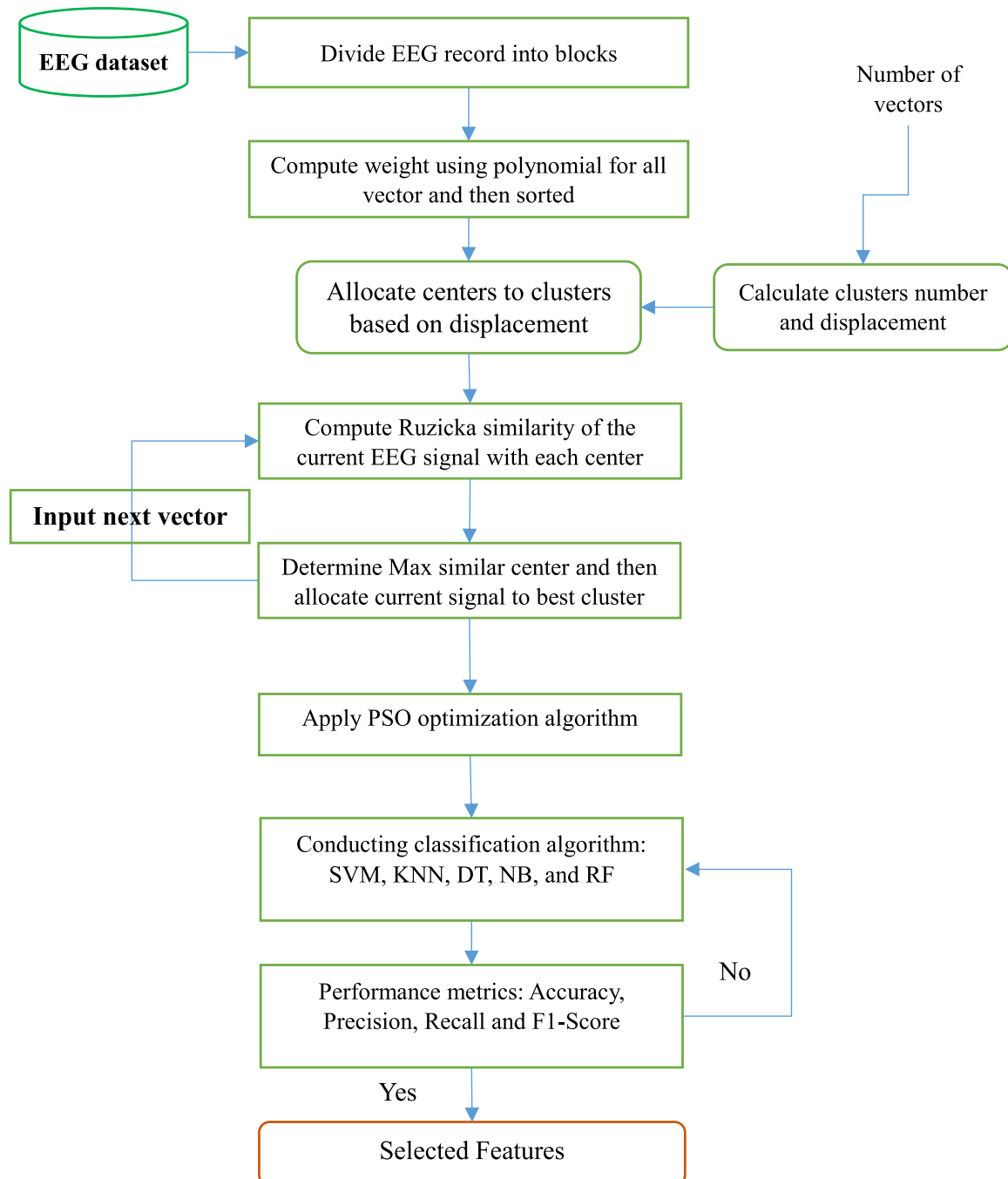


Fig. 2. Proposed Ruzicka Similarity-Based Clustering Model Integrated with PSO Algorithm.

required spacing for clustering. The polynomial equation is far superior to classical cluster center determination techniques, particularly in terms of simplicity. The cluster centers calculation is given by Eq. (2).

$$\mathbf{W} = \sum_{k=1}^n k * \mathbf{x}_k \quad (2)$$

where:  $n$  denotes the total number of items in the feature vector,  $\mathbf{x}_k$  represents the value of the  $k^{th}$  vector element,  $k$  is the vector index (ranging from 1 to  $n$ ), and  $\mathbf{W}$  is the resulting weight assigned to the feature vector. The function of polynomial weighting is very vital in clustering with respect to initializing the cluster centers. It assigns all feature vectors a deterministic weight with respect to the combination of feature values and positional indices instead of relying upon random or heuristic-based initialization which keeps giving a different and unstable clustering output. This way, more influential vectors can be offered precedence with respect to center cluster selection.

Assign each EEG record to the cluster with which it has the highest similarity. For a record, Ruzicka similarity is calculated with both cluster centers, and then if the first center has a higher similarity, the record is allocated to the first cluster. Otherwise, if the second center yields higher similarity, the record is assigned to the second cluster. This means that each EEG signal will in effect be grouped according to its best resemblance to one of the cluster centers. For example, if a record has similarity scores of 0.82 for the first cluster and 0.65 for the second, the record will be placed in the first cluster. The similarity value's interpretation further directs the clustering process. A Ruzicka value close to 1.0 indicates that there is near-identity for two vectors, while a value near 0 indicates minimal similarity. Values higher than 0.8 may be considered a strong match in the sense that EEG signals belonging to the same cluster share meaningful structural characteristics. However, the 0.8 threshold remains heuristic without data-driven calibration. The proposed approach offers stable and efficient clustering due to the combination of polynomial-based cluster center initialization with Ruzicka similarity, which in turn serves as a solid basis for PSO-based feature selection and seizure detection.

As illustrated in Fig. 2, the first process in proposed clustering model is reading EEG dataset, which serves as input. Then, EEG data is divided into blocks for processing. After that, for each vector in EEG blocks, compute weight based on polynomial function and then sorting these weights. Next, calculate the number of clusters and allocate center to these clusters. After that, compute Ruzicka similarity metric to measure the similarity between the cluster centers and current EEG signal. PSO optimization algorithm will be applied to refine the clustering or feature selection process. Then the selected features will be fed to different ML classifiers such as RF, NB, SVM, KNN, and DT. Then, ML classifiers are evaluated based on the performance evaluation metrics of accuracy, recall, precision, and F1-score.

#### 4. Experiments and results

All experiments were performed using the Bonn EEG dataset, on which the proposed Ruzicka-PSO clustering framework was tested with five ML algorithms: RF, NB, SVM, KNN, and DT. The evaluation of the models was based on an 80:20 division of the entire data into a training set (for training purposes) and a test set (to test the performance of the classification algorithms). The experimental pipeline followed EEG clustering against the Ruzicka similarity metric, then feature selection from the various clustered sets using a PSO optimization algorithm. The selected features are passed to the classifiers, performance evaluation is carried out with Accuracy, Precision, Recall, and F1 score as the primary measures for performance evaluation. The main hyperparameters of the PSO algorithm and the classifiers are summarized in Table 2. EEG records were used in their raw form without any preprocessing or filtering. No normalization and no artifact removal were applied. This decision allows the performance of the proposed Ruzicka-PSO framework to be

**Table 2**

Key Hyperparameters of PSO and Classifiers Used in the Experiments.

| Method | Hyperparameters                                                                                    |
|--------|----------------------------------------------------------------------------------------------------|
| PSO    | Swarm size = 30, inertia weight = 0.7, acceleration coefficients $c1 = c2 = 1.5$ , iterations = 30 |
| SVM    | Kernel = Radial Basis Function (RBF), $C = 1.0$                                                    |
| KNN    | $k = 5$ , distance = Euclidean                                                                     |
| DT     | Criterion = Gini, max depth = 10                                                                   |
| RF     | 100 trees, max depth = 15                                                                          |
| NB     | Gaussian NB, default variance smoothing                                                            |

validated directly on EEG signals.

##### 4.1. Dataset description

The present study used the EEG data set from the University of Bonn for the experiment [25], which is a benchmark dataset for seizure detection. The data comprises 500 recordings: each recording represents an EEG segment of 23.6 s duration sampled at 173.6 Hz with 4097 data points (features). These signals are divided into 5 subsets (O, Z, N, F, and S), with each subset comprising 100 recordings. The chaotic properties of the EEG signals were defined through the use of correlation dimension and nonlinear prediction-error based on some nonlinear time series methods. Surrogate data testing and weak stationarity analyses were then used to verify the stability of the segments. In Table 3, a concise description of the dataset configuration is presented.

##### 4.2. Evaluation metrics

In this study, the effectiveness of the proposed Ruzicka-PSO clustering framework is evaluated on the bases of Accuracy, Precision, Recall, and F1-score. These metrics are primary evaluation metrics in ML, as they provide the basis for validating classification models in a standardized and interpretable form across various domains, including biomedical signal analysis [26,27]. Statistical significance testing is useful in comparative scenarios. However, these tests rely on certain assumptions (such as paired predictions on a specific test set). Hence, we emphasize the importance of these four metrics that predominantly describe the reliability that applies to any other metrics used in judging the effectiveness of the proposed method.

A True Positive (TP) identifies seizure recordings identified as seizures. A True Negative (TN) identifies healthy recordings identified as being healthy. A False Positive (FP) is when healthy recordings are incorrectly classified as seizures. A False Negative (FN) is when seizure recordings are incorrectly classified as healthy [28]. Based on these definitions, the four primary metrics are:

● Accuracy: the proportion of EEG recordings (both seizure and healthy) that are correctly classified out of the total number of recordings, as expressed in Eq. (3).

$$\text{Accuracy} = \frac{TN + TP}{FP + FN + TN + TP} \quad (3)$$

● Precision: Among the recordings identified as seizures by the model, the proportion that are truly seizure cases, as described in Eq. (4).

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

● Recall: Among all actual seizure recordings, the proportion that the model successfully detects as seizures, as shown in Eq. (5).

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

● F1-score: A single measure that balances Precision and Recall, particularly useful when seizure and healthy recordings are unevenly

**Table 3**

Description of the Bonn EEG Dataset.

| Set | Status                                        | Features | Recordings | Duration (s) | Patients | Sampling Rate (Hz) |
|-----|-----------------------------------------------|----------|------------|--------------|----------|--------------------|
| O   | Healthy (eyes closed)                         | 4097     | 100        | 23.6         | 5        | 173.6              |
| Z   | Healthy (eyes open)                           | 4097     | 100        | 23.6         | 5        | 173.6              |
| N   | Epileptic, seizure-free (epileptogenic zone)  | 4097     | 100        | 23.6         | 5        | 173.6              |
| F   | Epileptic, seizure-free (opposite hemisphere) | 4097     | 100        | 23.6         | 5        | 173.6              |
| S   | Epileptic, ictal (during seizure)             | 4097     | 100        | 23.6         | 5        | 173.6              |

distributed, as described in Eq. (6).

$$F1 - \text{Score} = \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \times 2 \quad (6)$$

In this study, we did not conduct any statistical significance testing (e.g., ANOVA or Friedman). It follows that all reported results are to be interpreted in a descriptive manner rather than as statistically validated comparisons. In the evaluation, different subsets of these recordings were incorporated into multi-class and binary classification tasks, as summarized in Table 4. The proposed test scenarios address simple tasks such as distinguishing seizure from healthy or the exceedingly complex that involve multiple conditions. A total of 11 test scenarios were conducted in the experimental setup, thereby ensuring a thorough assessment of the model's performance in diverse clinical conditions.

#### 4.3. Performance of ML classifiers with the proposed Ruzicka-PSO method

Table 5 presents the outcomes of accuracy that show how the proposed method is efficient. The representation of this is done in a more pictorial form in Fig. 3. Whenever the five classifiers recorded (RF, NB, DT, KNN, and SVM) were tested, it was discovered that NB often performed best consistently and it reached 100 % at sometimes across most class combinations. RF was always performing on top among all, achieving an accuracy of 100 % in many cases and around 70 % in almost all situations. DT has done very well, though in some cases it has failed with many classes. Although SVM accuracy decreases with more class combinations, it has proven to be more effective than other models in similar categories. The accuracy of KNN starts from 63.33 % to 43.75 %, and its results seem to be lower than those of other classifiers.

The various classifiers of ML (RF, NB, DT, KNN, and SVM) show a high percentage of accuracy in terms of using PSO + Ruzicka clustering approach as indicated in Fig. 3. RF consistently achieves the highest accuracy, followed by SVM and Decision Tree, particularly with

**Table 4**

Evaluation Strategy and EEG Test Scenarios.

| Test number | Sets of EEG signals       | Description                                                                                    |
|-------------|---------------------------|------------------------------------------------------------------------------------------------|
| Test 1      | Z vs. S                   | Healthy (eyes open) vs. Epileptic seizure                                                      |
| Test 2      | O vs. S                   | Healthy (eyes closed) vs. Epileptic seizure                                                    |
| Test 3      | F vs. S                   | Seizure-free (epileptogenic zone) vs. Epileptic seizure                                        |
| Test 4      | N vs. S                   | Seizure-free (opposite hemisphere) vs. Epileptic seizure                                       |
| Test 5      | Z vs. O vs. S             | Two healthy conditions (eyes open/closed) vs. Epileptic seizure                                |
| Test 6      | F vs. S vs. N             | Seizure-free (epileptogenic/opposite hemisphere) vs. Epileptic seizure                         |
| Test 7      | O vs. Z vs. N vs. S       | Healthy (eyes open/closed), seizure-free (opposite hemisphere), and Epileptic seizure          |
| Test 8      | O vs. Z vs. F vs. S       | Healthy (eyes open/closed), seizure-free (epileptogenic zone), and Epileptic seizure           |
| Test 9      | F vs. N vs. O vs. S       | Seizure-free (epileptogenic/opposite hemisphere), healthy (eyes closed), and Epileptic seizure |
| Test 10     | F vs. N vs. Z vs. S       | Seizure-free (epileptogenic/opposite hemisphere), healthy (eyes open), and Epileptic seizure   |
| Test 11     | N vs. F vs. O vs. Z vs. S | All categories combined                                                                        |

**Table 5**

Classification Accuracy of RF, NB, DT, KNN, and SVM using Ruzicka Similarity-based Clustering With PSO on EEG Signals (4097 Features, 23.6 s Duration, 20 % Testing Size).

| Classes           | SVM   | KNN   | DT    | NB    | RF    |
|-------------------|-------|-------|-------|-------|-------|
| S - Z             | 93.33 | 63.33 | 93.33 | 100   | 100   |
| S - O             | 90.32 | 61.29 | 93.54 | 100   | 96.77 |
| S - F             | 90    | 56.66 | 70    | 96.66 | 100   |
| S - N             | 92.30 | 92.30 | 92.30 | 100   | 92.30 |
| O - Z - S         | 70.45 | 50    | 65.90 | 90.90 | 90.90 |
| F - N - Z         | 69.44 | 50    | 72.22 | 72.22 | 83.33 |
| O - Z - N - S     | 62.26 | 50.94 | 52.83 | 71.69 | 77.35 |
| O - Z - S - F     | 54.54 | 43.63 | 56.36 | 69.09 | 65.45 |
| F - S - O - N     | 58.82 | 52.94 | 62.74 | 64.70 | 86.27 |
| F - S - Z - N     | 63.63 | 57.57 | 57.57 | 42.42 | 60.60 |
| N - F - O - S - Z | 48.43 | 43.75 | 40.62 | 60.93 | 71.87 |

essential feature selection. Classifier performance changes as feature complexity increases, with more features not always resulting in improved accuracy. Overall, PSO + Ruzicka significantly enhances classification, producing better results than other classification methods. Figs. 4–6 explain precision, recall, and F1-score metrics using Ruzicka similarity based on clustering model for PSO.

#### 4.4. Performance of ML classifiers with standard PSO feature selection

Several models, including DT SVM, NB, KNN, and RF, are employed to compare the performance of the suggested method with PSO algorithm. The proposed model exhibited effectiveness and robustness in the classification process compared to PSO, and it surpassed PSO without clustering support in all instances (binary and multi-class combinations) using SVM, DT, KNN, NB, and RF. Experiments with RF classifier yielded an accuracy ranging from 96.77 % to 100 %. However, five classes combining (Z, O, F, N, and S) and applying RF classification algorithm, the model's accuracy lagged significantly. The proposed technique achieved an accuracy of 71.87 %, while PSO alone reached 73 % with the same classifiers and class combinations. Table 6 which is presented the accuracy of the results obtained using PSO algorithm.

Table 6 compares the accuracy of DT, SVM, NB, RF and KNN classifiers for PSO optimization algorithm of EEG data (23.6 s, 4097 features, 20 % testing size) without clustering. RF consistently obtains the maximum accuracy, especially in smaller feature sets such as Z,S (100 %) and N,S (100 %), but SVM performs poorly in all circumstances. A moderate degree of accuracy is achieved by DT, with the best results coming from simpler sets and diminishing accuracy with increasing combinations. In spite of its competitive performance, NB has trouble under highly loaded conditions. Overall, fewer feature sets lead to higher accuracy, with RF emerging as the most reliable classifier.

Fig. 7 explains the results of accuracy for various classifiers DT, RF, NB, SVM, and KNN using the optimizer method PSO based on EEG data (23.6 s, 4097 characteristics). RF has the highest obtained accuracy across several feature sets, followed by NB and DT. SVM performs poorly in most circumstances, with much-reduced accuracy. Although testing classes (S, Z; S, O; S, F; S, N) achieved better accuracy, the combinations (O, Z, S, F; N, F, O, S, Z) caused a substantial performance loss for all classifiers. The findings show that PSO optimization increases classification accuracy, with RF emerging as the most robust model.

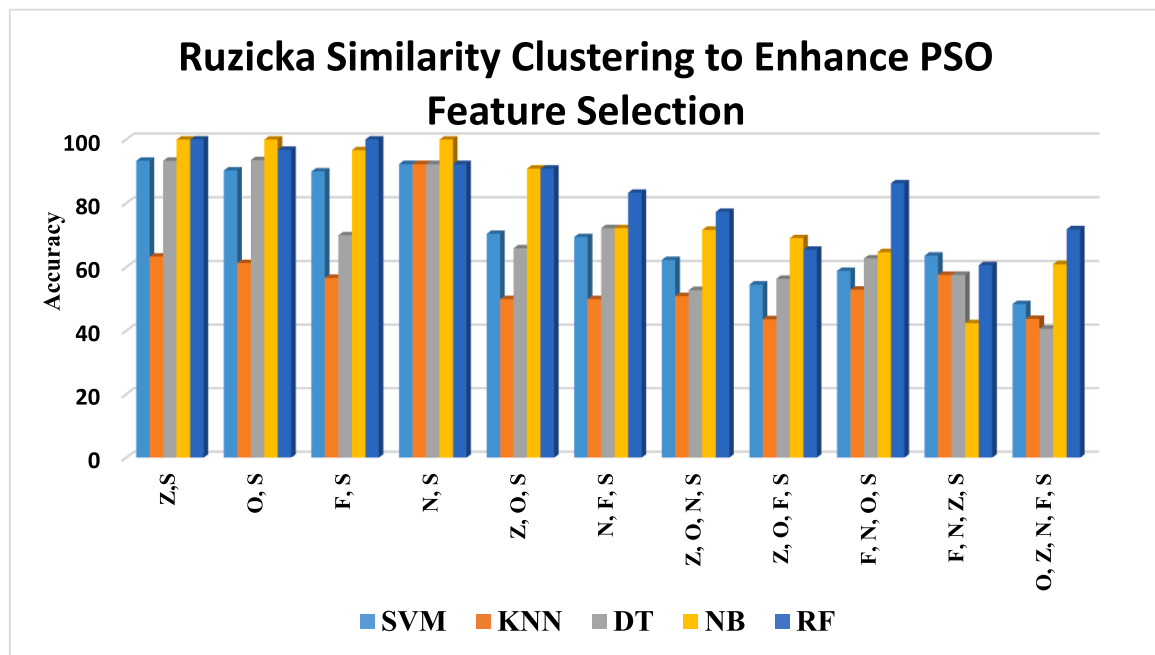


Fig. 3. Classification Accuracy Outcomes Using Ruzicka Similarity-Based Clustering with PSO on EEG Signals (4097 Features, 23.6 s Duration).

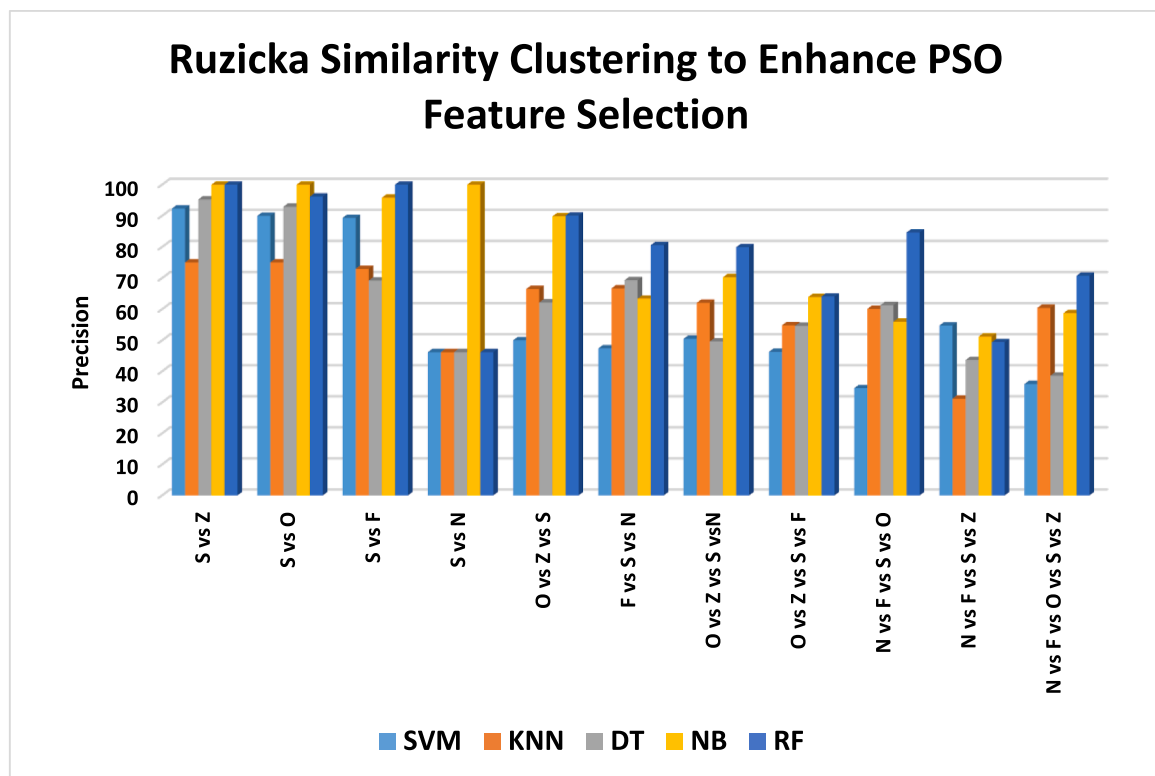


Fig. 4. Precision Outcomes Using Ruzicka Similarity-Based Clustering with PSO on EEG Signals (4097 Features, 23.6 s Duration).

#### 4.5. Discussion of results

This section describes the experimental findings of the proposed framework, evaluating its merits and shortcomings, to appreciate its performance in EEG-based seizure detection. The most prominent findings are summarized as follows:

- **Multi-class classification performance:** In binary tasks, the proposed method performed with high accuracy. However, the performance significantly deteriorated under multi-class settings, achieving an accuracy range of 48.43 %–90.90 % depending on the class combinations and classifiers used. For instance, in the O–Z–S experiment, classification accuracy with NB and RF reached 90.90 %, while prediction with KNN reduced to 50 %. In the most complex five-class test (N–F–O–S–Z), the highest performance was achieved by RF with

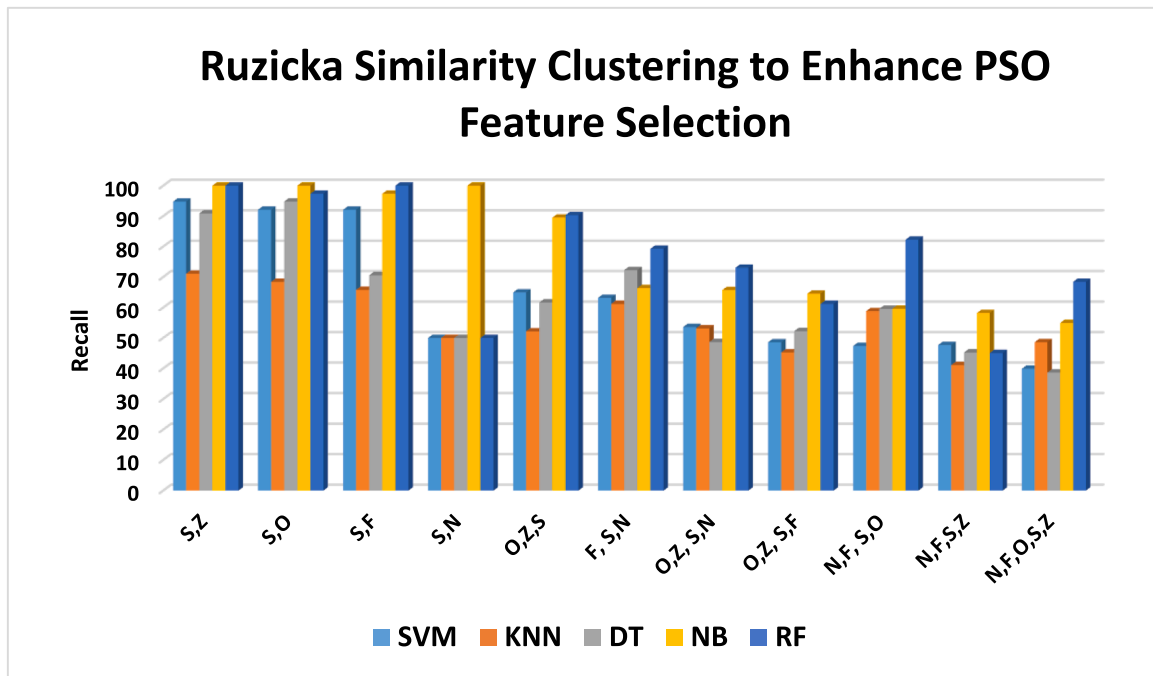


Fig. 5. Recall Outcomes Using Ruzicka Similarity-Based Clustering with PSO on EEG Signals (4097 Features, 23.6 s Duration).

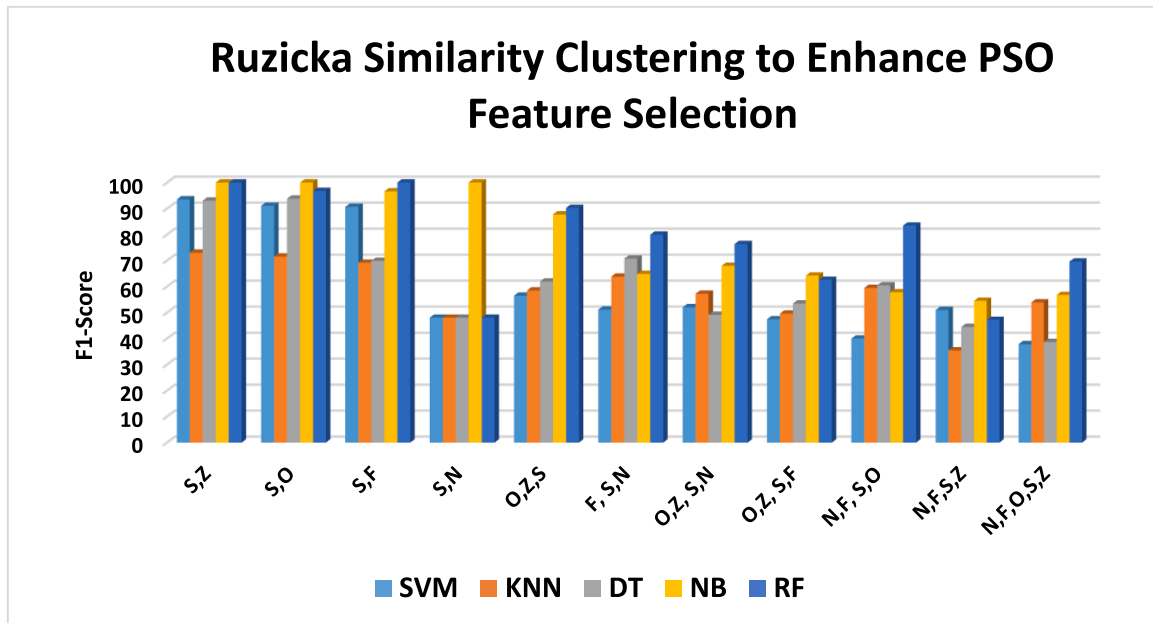


Fig. 6. F1-Score Outcomes Using Ruzicka Similarity-Based Clustering with PSO on EEG Signals (4097 Features, 23.6 s Duration).

71.87 %, with other classifiers scoring lower than this. Compared to PSO without clustering, which recorded significantly weaker results (as low as 17–65 % in some tests), Ruzicka-PSO approach clearly improves robustness. Nonetheless, the reduced performance in multi-class cases can be attributed to overlapping EEG features across classes, subtle variations in class distributions, and the fixed two-cluster design, which may not fully capture the complexity of datasets with multiple classes. Several factors contribute to the reduced performance in multi-class classification. First, EEG signals often exhibit overlapping features between different brain states, making it difficult for classifiers to clearly separate seizure-free, ictal, and healthy recordings when more than two classes are involved.

Second, Bonn dataset, although balanced in terms of the number of recordings per class, may still present subtle class imbalance in feature distributions, which can bias decision boundaries in multi-class setups.

- Computational efficiency considerations: PSO works basically on the entire high-dimensional feature space, making it more computationally intensive and prone to slow convergence. EEG records are first clustered with the Ruzicka similarity measure in order to partition the feature space into a number of more homogeneous subsets. Consequently, this reduces the optimization search complexity of the PSO since it will only be exploring a more compact and harmonized space within each cluster. In addition, clustering

**Table 6**

Classification Accuracy of RF, NB, DT, KNN, and SVM using PSO-based Feature Selection Without Clustering on EEG Signals (4097 Features, 23.6 s Duration, 20 % Testing Size).

| Classes           | SVM   | KNN   | DT    | NB    | RF    |
|-------------------|-------|-------|-------|-------|-------|
| S - Z             | 45    | 60    | 85    | 100   | 100   |
| S - O             | 45    | 47.5  | 72.5  | 97.5  | 95    |
| S - F             | 45    | 47.5  | 75    | 92.5  | 95    |
| S - N             | 45    | 47.5  | 85    | 95    | 100   |
| O - Z - S         | 33.33 | 38.33 | 56.66 | 76.66 | 71.66 |
| F - N - Z         | 45    | 63.33 | 63.33 | 68.33 | 95    |
| O - Z - N - S     | 20    | 38.75 | 47.5  | 65    | 70    |
| O - Z - S - F     | 20    | 42.5  | 51.24 | 62.5  | 82.5  |
| F - S - O - N     | 20    | 47.5  | 51.24 | 57.49 | 76.25 |
| F - S - Z - N     | 20    | 50    | 48.75 | 48.75 | 73.75 |
| N - F - O - S - Z | 17    | 30    | 42    | 62    | 73    |

using the Ruzicka measure inhibits the PSO from performing unnecessary evaluations on weakly related features which is conducive toward hastening convergence by reducing the number of required iterations. Although exact runtime measurements were not reported, this qualitative reduction in search space complexity provides a strong rationale for expecting improved efficiency compared to PSO alone.

- Scalability for real-time applications: Although the proposed method demonstrates strong performance in offline experiments, its scalability for real-time EEG monitoring remains untested. In practical deployment, runtime and latency may be influenced by factors such as the sampling rate of the incoming EEG stream, the number of channels processed simultaneously, available memory bandwidth, and whether the Ruzicka-PSO pipeline is executed on a Central Processing Unit (CPU), Graphics Processing Unit (GPU), or edge hardware. However, because the clustering step reduces the search space for PSO and simplifies the feature selection stage, the proposed approach may support real-time scalability, although this has not yet been empirically evaluated.

#### 4.6. Comparison with existing techniques

A comparison of the proposed model with current methods is carried out using Bonn University dataset. In comparison with standard PSO and other techniques, Ruzicka similarity based on clustering performed better than standard PSO alone. Table 7 compares the proposed method with a variety of state-of-the-art approaches. Graphically, results are presented in Fig. 8.

Table 7 compares the accuracy of classifiers using various clustering algorithms, including the proposed PSO Ruzicka similarity clustering. Compared to prior studies, Minkowski-based PSO clustering outperformed NB (100 %) and RF (96.6 %), although SVM did badly (60 %). The PSO + FCM approach with LSTM obtained 98.5 % accuracy, whereas PSO + KPCA with a Deep Belief Network achieved 94 %. Other clustering techniques, such as Cuckoo Search, K-means, and Gower distance, achieved remarkable accuracy, especially SVM (99.48 %), NB+Linear SVM (98.96 %), and RF (98 %). The suggested clustering with Ruzicka Similarity and PSO feature selection performs admirably, attaining 100 % accuracy with both NB and RF and greatly enhancing SVM and DT (93.33 %). Findings show that the proposed method was very efficient as it achieved a maximum of 100 % with both RF and NB classifiers on EEG signals classification. Algorithm 1

#### 4.7. Limitations of the study

The proposed study has certain limitations that may restrict its applicability in broader contexts. The key limitations are:

- Single dataset: This study exclusively utilized the Bonn EEG dataset collected under controlled experimental conditions in a single-center setup. Although the most popular and reliable benchmark, this dataset may not account for the whole variability associated with real-world clinical settings. For this reason, its findings may thus be limited regarding generalizability.
- Sensitivity to incomplete or noisy data: The suggested approach assumes that the EEG recordings are entirely intact and clean. In reality, clinical datasets more often present with missing entries or

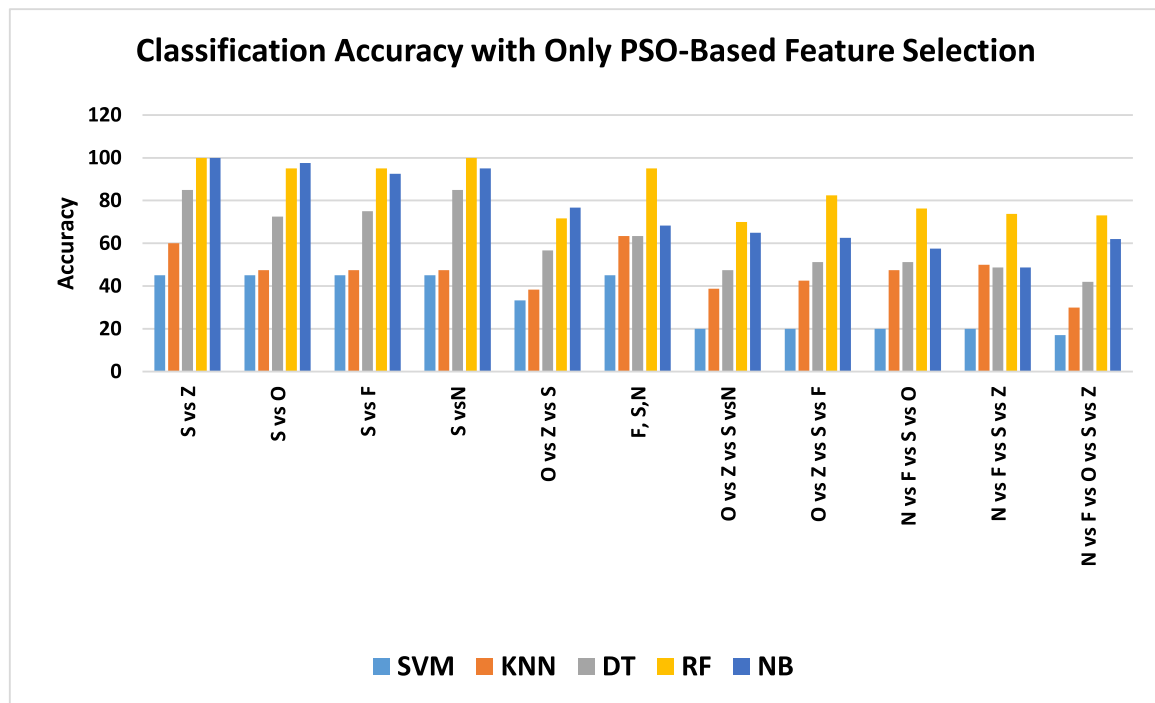


Fig. 7. Classification Accuracy Outcomes for PSO-Based Feature Selection Without Clustering on EEG Signals (4097 Features, 23.6 s Duration).

**Table 7**

Comparison of the Proposed Ruzicka Similarity-Based Clustering Algorithm with Recent Clustering Methods.

| Ref/year                                          | Suggested approach                                                                                     | Classifier                    | Accuracy (%) |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------|--------------|
| Al-Shammary et al. [20] (2024)                    | PSO + Clustering approach based on Minkowski                                                           | DT                            | 75           |
|                                                   |                                                                                                        | RF                            | 96.6         |
|                                                   |                                                                                                        | SVM                           | 60           |
|                                                   |                                                                                                        | NB                            | 100          |
|                                                   |                                                                                                        | KNN                           | 53           |
| Sunil Kumar Prabhakar and Dong-Ok Won [21] (2023) | Cuckoo search cluster<br>K-mean cluster<br>Fuzzy C-mean (FCM)<br>Dragonfly cluster<br>Firefly clusters | SVM                           | 99.48        |
|                                                   |                                                                                                        | NB+Linear SVM                 | 98.96        |
|                                                   |                                                                                                        | DT                            | 98.96        |
|                                                   |                                                                                                        | KNN                           | 75           |
|                                                   |                                                                                                        | NB                            | 75.75        |
| Kamakshi and Rengaraj [29] (2024)                 | PSO + Fuzzy c-means (FCM)                                                                              | Long short-term memory (LSTM) | 98.5         |
| Sharmila and Angel [30] (2024)                    | Kernel PCA + PSO                                                                                       | Deep belief network           | 94           |
| Baawi et al. [31] (2025)                          | Gower distance                                                                                         | RF                            | 98           |
| <b>Proposed</b>                                   | <b>Clustering with Ruzicka similarity and PSO feature selection</b>                                    | <b>SVM</b>                    | <b>93.33</b> |
|                                                   |                                                                                                        | <b>KNN</b>                    | <b>63.33</b> |
|                                                   |                                                                                                        | <b>DT</b>                     | <b>93.33</b> |
|                                                   |                                                                                                        | <b>NB</b>                     | <b>100</b>   |
|                                                   |                                                                                                        | <b>RF</b>                     | <b>100</b>   |

corrupted segments due to artifacts and interruptions in the recording. The presence of such interruptions may indeed affect the model's ability to identify suitable cluster centers and consequently lower the reliability of the whole clustering process.

- Fixed number of clusters: The EEG records in this work have been clustered into two clusters. This may, however, not hold for other EEG datasets which contain more than two or three classes. A fixed number of the method may make it inflexible to deal with more complex or heterogeneous structures of datasets.

#### 4.8. Future work

For future works, several directions can be pursued to enhance the robustness and generalization of the model:

- Dynamic clustering: Future research can focus on developing adaptive clustering techniques that automatically adjust the number of clusters according to the dataset's characteristics. Such a framework would have better flexibility and could be used with EEG datasets with either multiple classes or heterogeneous distributions instead of just fitting into a fixed number of clusters.
- Hybrid feature selection: While PSO provides effective feature selection, extending the approach by combining filter-based and wrapper-based methods may yield more optimal subsets of features. Such hybrid strategies could balance computational efficiency with discriminative power, leading to more reliable classification outcomes across different datasets.
- Alternative methods for cluster center identification: The current study employs a polynomial-based approach to determine cluster centers. Future studies could explore other strategies, such as density-based center selection, medoid-based clustering, or optimization-driven approaches, to improve the stability and accuracy of cluster initialization.
- Real-time performance evaluation: Future work will include profiling runtime, memory usage, and latency during continuous EEG stream processing to assess whether the model can meet the timing and responsiveness requirements necessary for real-time monitoring environments.

#### 5. Conclusion

The paper proposed efficient cluster technique founded on Ruzicka similarity for improving EEG signal classification efficiency. The proposed method begins from the EEG dataset, which is divided into two contrasting clusters C1 and C2 utilizing the Ruzicka similarity measure. This clustering process guarantees grouping of data points with high similarities, thus facilitating the structure of the dataset. After the

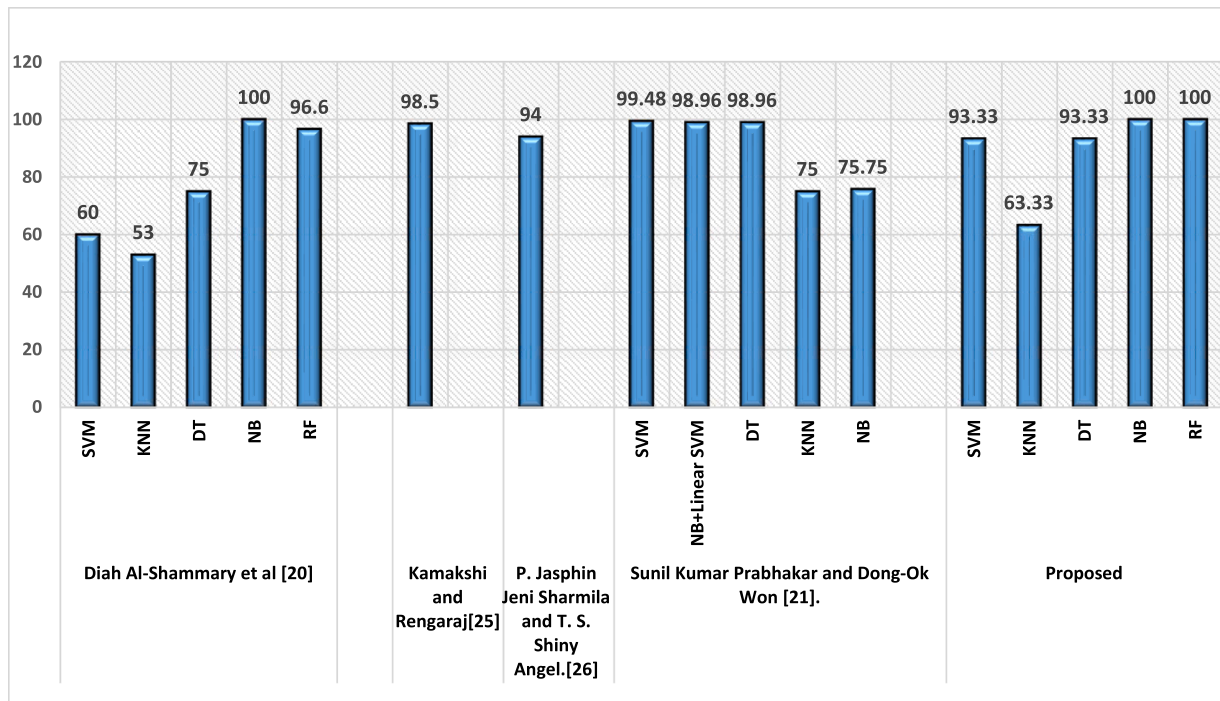


Fig. 8. Comparison of the Proposed Ruzicka-PSO Model with Recent Clustering Methods.

**Algorithm 1**

Ruzicka Feature Selection based on Clustering Model for PSO.

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**Input:** S\_No // It represents the number of vector in the dataset  
 C\_No // It refers to the number of clusters  
 V // It denotes vector items  
 All \_ polynomial\_np.argsort(S\_No) // Array of Numpy to determine the allocated center

**Output:**  
 Cluster\_0\_with\_labels // A CSV file containing the results of Cluster 1, with labels. Cluster\_1\_with\_labels // A CSV file containing the results of Cluster 2, with labels.

Steps: ● For each vector ( $i = 1$  to S\_No)    Begin    For each item in vector ( $j = 1$  to V)    Begin    Calculate weights    Compute the summation weights    Endfor ● For each vector item ( $i = 1$  to S\_No)    Begin

For each center in cluster ( $j = 1$  to C\_No)  
 Begin  
 Calculate the Ruzicka similarity between the current signal and each cluster center in the entire row.

$$\text{Ruzicka similarity} = \frac{\sum_{i=1}^{S\_No} \min \{ \text{current signal}_i, \text{currentcenter}_j \}}{\sum_{i=1}^{S\_No} \max \{ \text{current signal}_i, \text{currentcenter}_j \}}$$

Find the maximum similarity value.    Endfor    Add Vectors to Clusters    Endfor ● For each cluster ( $j = 1$  to C\_No)    Begin    Create a file (\*.csv) containing the allocated vectors for that cluster.    Endfor ● Return file

---

clustering process has been completed, the application of PSO then follows, in order to search for high-convergence features within each of the clusters. This is the most effective way where PSO chooses the more optimal features, reducing the dimensional space, but retaining important information. The selected features are afterwards used in training traditional classifiers such as DT, SVM, NB, KNN, and RF. The performance evaluation is done based on accuracy, precision, recall, and F1-score metrics. The method proposed achieves a classification accuracy of up to 100 %, which proves the effectiveness of diagnosis of epilepsy compared to standard approaches. Future work can be focused on improving feature selection by employing other optimization techniques like Artificial Bee Colony (ABC) and Grey Wolf Optimization (GWO). Additionally, incorporating preprocessing techniques such as DWT may help reduce noise and improve signal quality. Expanding the evaluation to include more EEG datasets can also validate the robustness and generalizability of the proposed model.

**Funding**

Sponsored by Victoria University, Australia.

**Ethics statement**

There are no ethical issues involved in this research.

**CRedit authorship contribution statement**

**Sarah L. Alzamili:** Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Salwa Shakir Baawi:** Writing – review & editing, Visualization, Supervision, Methodology, Formal analysis, Conceptualization. **Mustafa Noaman Kadhim:** Writing – review & editing, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Dhiah Al-Shammary:** Writing – review & editing, Validation, Supervision, Software, Methodology, Formal analysis, Conceptualization. **Ayman Ibaida:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Methodology, Data curation, Conceptualization. **Khandakar Ahmed:** Writing – review & editing, Validation, Supervision, Resources, Formal analysis, Conceptualization.

**Declaration of competing interest**

No conflict of interest exists.

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

**Data availability**

The EEG Bonn University dataset used in this study is publicly available and can be accessed at (<https://www.ukbonn.de/epileptologie/arbeitsgruppen/ag-lehnertz-neurophysik/downloads/>).

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